CSAI 498 / CSAI 499 – Graduation Project Proposal

Project Title:

AI-Powered Smart Campus Assistant for Zewail University

Team Members:

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Abstract

This project aims to develop an AI-Powered Smart Campus Assistant for Zewail University for Science and Technology. The system will serve as an intelligent digital assistant that supports students in academic life efficiently. It will answer questions related to class schedules, faculty offices, and university policies using Retrieval-Augmented Generation (RAG) integrated with a Large Language Model (LLM). Additionally, it will guide users to lecture halls, offices, and libraries, as well as Personalized Study Recommendations based on user profiles and performance data. By leveraging AI and modern data technologies, this assistant will enhance student experience, reduce administrative load, and establish a scalable framework that could later be deployed on the university's official website. The expected outcome is an integrated smart system that improves campus accessibility, communication, and personalized academic support.

Problem Statement & Motivation (SO 1)

Students at Zewail University often face challenges in accessing campus information efficiently — such as finding lecture halls, checking updated schedules, or identifying faculty office locations. Currently, students rely on scattered resources like emails, static portals, and manual directions. These fragmented systems can waste time and lead to confusion, especially for new students.

In addition, many students experience difficulties due to the unavailability of the university's self-service system sometimes because of account holds, or delays in registration, which prevent them from accessing critical academic information. New students, in particular, often do not thoroughly read the student handbook and thus have numerous inquiries about university policies, course descriptions, credit hours, and prerequisites. These recurring issues result in a high volume of routine questions directed toward administrative staff and academic advisors, slowing down both student and staff workflows.

The lack of a unified, intelligent information hub impacts academic productivity and campus experience. Therefore, an AI-driven assistant that can understand natural language queries and provide context-aware responses would greatly enhance efficiency. This problem is significant because it addresses daily student needs, streamlines administrative communication, and aligns with Zewail University's vision of digital transformation and technological excellence.

Proposed Solution (SO 1 & SO 2)

The proposed AI-Powered Smart Campus Assistant will integrate Retrieval-Augmented Generation (RAG) with a Large Language Model (LLM) to answer questions based on university data and policies. The system will provide:

1. Automated Q&A about schedules, courses, and university regulations.
2. Information lookup for rooms and offices.
3. Personalized Study Recommendations based on academic performance.

The innovation lies in combining AI-driven natural language understanding, campus-specific data retrieval, and spatial intelligence within one platform. It bridges multiple technologies to create a context-aware assistant aligned with real university needs.

Project Scope (SO 2)

In Scope (Must-have):

- RAG-based Q&A connected to the university database.
- LLM integration for natural language understanding.
- Personalized study recommendation engine.
- Web/Mobile user interface.

Could-have (Optional Features):

- Voice assistant capabilities.
- Integration with academic calendars.
- Multilingual support (Arabic/English).

Out of Scope (Will-not-have):

- GPS-based navigation.
- Integration with third-party platforms.

Assumptions / Limitations:

- Access to Zewail University’s data will be granted.

High-Level Timeline (SO 2)

Phase	Description	Duration (weeks)	Deliverables
Research & Requirement Analysis	Gather data, define requirements, explore AI systems.	3	Requirement document, architecture draft
Design & Planning	Design architecture, choose frameworks, prepare dataset pipelines.	3	System design document, mockups
Implementation Part 1	Develop backend, RAG, LLM integration, and frontend prototype.	6	Working prototype
Testing & Evaluation	Test system accuracy, QA performance, and usability.	3	Final system, evaluation report

Technology Stack & Theoretical Basis (SO 6 – DSAI Focus)

Programming Languages: Python, JavaScript (React / Flutter)
Frameworks & Libraries: PyTorch, Hugging Face Transformers, LangChain, FastAPI, React Native
Databases: PostgreSQL, Pinecone / FAISS
APIs & Tools: OpenStreetMap, OpenAI API

Justification:
- Python for strong AI/NLP support.
- LangChain + RAG for efficient document retrieval.
- FastAPI + React for scalable architecture.
- Vector Databases for fast semantic search.

Program Focus (DSAI): Emphasizes AI lifecycle management, data retrieval optimization, and evaluation of LLM-based systems.

Success Metrics & Evaluation Plan (SO 2 & SO 6)

Success Metrics:
- Response Accuracy: ≥ 90% relevance.
- User Satisfaction: ≥ 85% positive feedback.
- Performance: Response time ≤ 5 seconds.

Evaluation Plan:
- Internal testing with campus-specific queries.
- Compare AI vs human answers.
- User testing with students.
- Log performance metrics.

Team Roles & Responsibilities (SO 5 & SO 3)

Team Member	Role	Responsibilities
Ahmed Mongi	Data & Backend	Manage RAG pipeline, handle database.
Omar Mohamed	AI/NLP Integration	Fine-tune and integrate LLM models.
Joumana Mohamed	Web Developer	Design and build frontend interface and backend APIs.
Sama Magdy	Data Integration	Manage campus-related information and APIs.

Communication Plan:

- Weekly team meetings with the supervisor.
- Daily coordination via Slack/Discord.
- Documentation and tracking on GitHub.

References

1. Y. Liu et al., “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” arXiv:2005.11401, 2020.
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3. LangChain Documentation – https://www.langchain.com/
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